arXiv:2505.08532v1 [cs.SI] 13 May 2025

真理越辩越明！基于大语言模型的多智能体系统揭露假新闻基于大语言模型的多智能体系统揭露假新闻

刘宇寒 北京大学人工智能学院 中国人民大学 北京，中国
y
uhan.liu@ruc.edu.cn

刘宇轩 北京大学人工智能学院 中国人民大学 北京，中国

张晓晴 北京大学人工智能学院 中国人民大学 北京，中国

陈秀莹*阿联酋扎耶德人工智能大学 阿布扎比，阿联酋

瑞安*
北京航空航天大学，中国中国人民大学，北京，中国武汉大学，武汉，中国

摘要

在当今的数字环境中，假新闻通过社交网络快速传播，对社会构成了重大挑战。现有的检测方法大多采用传统的分类模型，这些模型存在可解释性低和泛化能力有限的问题；或者为大型语言模型（LLM）设计特定提示，直接生成解释和结果，未能充分利用LLM的推理能力。受“真理越辩越明”这句话的启发，我们的研究引入了一种名为TruEDebate（TED）的基于LLM的多智能体系统，以增强假新闻检测的<code style id='1'>可解释性</code>和<code style id='3'>有效性</code>。TED采用了一种受正式辩论场景启发的严格辩论过程。我们方法的核心是两个创新组件：DebateFlow代理和InsightFlow代理。DebateFlow代理将代理组织成两个团队，一个支持新闻的真实性，另一个挑战其真实性。这些代理进行开场陈述、交叉询问、反驳和总结陈词，模拟类似于人类话语分析的严格辩论过程，从而对新闻内容进行全面评估。同时，InsightFlow代理由两个专门子代理组成：合成代理和分析代理。合成代理总结辩论并提供总体观点，确保评估的连贯性和综合性。分析代理包括一个角色感知编码器和一个辩论图，它将角色嵌入和模型集成在一起通过注意力机制，探讨辩论角色和论点之间的交互，并提供最终判断。我们在ARG-EN和ARG-CN两个数据集上的大量实验表明，TED框架在各项指标上超越了传统方法，更重要的是，通过揭示逻辑推理和结构化辩论过程，增强了可解释的假新闻检测，从而得出准确结论。我们发布代码以支持在负责任信息系统中使用结构化辩论以改进决策[1]。

CCS 概念

• **计算方法** → **自然语言处理**；• **安全和隐私** → **安全和隐私的社会及人类方面**。

关键词

假新闻，辩论，大语言模型，多智能体系统

ACM 参考格式：

刘宇涵，刘宇轩，张晓庆，陈秀英和闫瑞。2025。辩论使真相更加清晰！ 基于大语言模型的多智能体系统揭露假新闻。收录于第*48*届国际*ACM SIGIR*信息检索研究与发展会议（*SIGIR '25*），*2025*年*7*月*13*日至*18*日，帕多瓦，意大利。ACM，纽约，纽约，美国，11页。https://doi.org/10.1145/3726302.3730092

1 简介

通过社交网络快速传播的虚假新闻已成为普遍现象，对个人、弱势群体和社会产生了重大影响。这种广泛的传播对公共舆论治理 [1]、影响选举 [11]、削弱公共卫生政策 [13]、公平性 [57–59]以及威胁社会稳定 [40]构成了相当大的挑战。因此，开发有效且可解释的虚假新闻检测方法的需求从未如此迫切 [37]。

*通讯作者。

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ACM
ISBN 979-8-4007-1592-1/2025/07
https://doi.org/10.1145/3726302.3730092

[1]https://github.com/LiuYuHan31/TED_虚假新闻检测

arXiv:2505.08532v1 [cs.SI] 13 May 2025

The Truth Becomes Clearer Through Debate! Multi-Agent Systems with Large Language Models Unmask Fake News

Yuhan Liu
Gaoling School of Artificial Intelligence,
Renmin University of China
Beijing, China
yuhan.liu@ruc.edu.cn

Yuxuan Liu
Gaoling School of Artificial Intelligence,
Renmin University of China
Beijing, China

Xiaoqing Zhang
Gaoling School of Artificial Intelligence,
Renmin University of China
Beijing, China

Xiuying Chen*
Mohamed bin Zayed University of Artificial Intelligence
Abu Dhabi, UAE

Rui Yan*
Gaoling School of Artificial Intelligence,
Renmin University of China
Beijing, China
Wuhan University
Wuhan, China

Abstract

In today's digital environment, the rapid propagation of fake news via social networks poses significant social challenges. Most existing detection methods either employ traditional classification models, which suffer from low interpretability and limited generalization capabilities, or craft specific prompts for large language models (LLMs) to produce explanations and results directly, failing to leverage LLMs' reasoning abilities fully. Inspired by the saying that "truth becomes clearer through debate," our study introduces a novel multi-agent system with LLMs named TruEDebate (TED) to enhance the *interpretability* and *effectiveness* of fake news detection. TED employs a rigorous debate process inspired by formal debate settings. Central to our approach are two innovative components: the DebateFlow Agents and the InsightFlow Agents. The DebateFlow Agents organize agents into two teams, where one supports and the other challenges the truth of the news. These agents engage in opening statements, cross-examination, rebuttal, and closing statements, simulating a rigorous debate process akin to human discourse analysis, allowing for a thorough evaluation of news content. Concurrently, the InsightFlow Agents consist of two specialized sub-agents: the Synthesis Agent and the Analysis Agent. The Synthesis Agent summarizes the debates and provides an overarching viewpoint, ensuring a coherent and comprehensive evaluation. The Analysis Agent, which includes a role-aware encoder and a debate graph, integrates role embeddings and models the interactions between debate roles and arguments using an attention mechanism, providing the final judgment. Our extensive experiments on two datasets, ARG-EN and ARG-CN, demonstrate that the TED framework surpasses traditional methods across various metrics and, more importantly, enhances interpretable fake news detection by illuminating logical reasoning and structured debate processes leading to accurate conclusions. We release our code to support Information systems that use structured debate within *responsible* information systems for improved decision-making[1].

CCS Concepts

• **Computing methodologies** → **Natural language processing**; • **Security and privacy** → **Human and societal aspects of security and privacy**.

Keywords

Fake News, Debate, Large Language Models, Multi-Agent System

ACM Reference Format:

Yuhan Liu, Yuxuan Liu, Xiaoqing Zhang, Xiuying Chen, and Rui Yan. 2025. ***The Truth Becomes Clearer Through Debate!*** Multi-Agent Systems with Large Language Models Unmask Fake News. In *Proceedings of the 48th International ACM SIGIR Conference on Research and Development in Information Retrieval (SIGIR '25), July 13–18, 2025, Padua, Italy*. ACM, New York, NY, USA, 11 pages. https://doi.org/10.1145/3726302.3730092

1 Introduction

The rapid propagation of fake news through social networks has become pervasive, significantly impacting individuals, vulnerable communities, and society. This widespread dissemination poses considerable challenges for the governance of public opinion [1], influencing elections [11], undermining public health policies [13], fairness [57–59] and threatening social stability [40]. Consequently, the urgency to develop effective and interpretable methods for detecting fake news has never been greater [37].

*Corresponding authors.

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SIGIR '25, Padua, Italy
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ACM ISBN 979-8-4007-1592-1/2025/07
https://doi.org/10.1145/3726302.3730092

[1]https://github.com/LiuYuHan31/TED_fake-news-detction

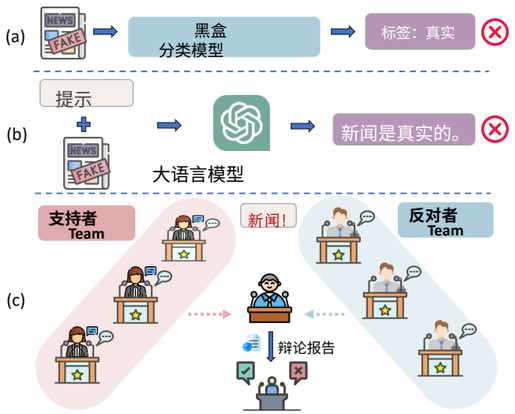


图1：(a) 传统模型主要依赖黑盒分类模型来检测假新闻。(b) 大语言模型被提示直接输出检测结果，这并未充分利用大型模型推理的潜力。(c) 我们的多人框架模拟了一个正式的辩论场景，就新闻主题进行讨论，通过辩论推理和可解释性来达成检测结果。

传统的假新闻检测方法主要依赖机器学习模型，特别是基于BERT [7]，的模型，这些模型在文本分析方面表现出色[16, 19, 31]。尽管这些模型取得了显著的成功，但它们通常存在可解释性低和泛化能力有限的问题，限制了它们在不同环境下的有效性。最近的研究尝试通过开发特定的提示来利用大语言模型，以生成解释和检测结果 [15, 29]。然而，这些方法可能未能充分利用大语言模型固有的推理能力和复杂的认知过程，经常导致结论过于简单化和草率。

在真理探索的背景下，谚语"真理越辩越明"强调了从多个视角审视和挑战信息以获得更准确理解的重要性 [2, 21]。结构化辩论中不同观点的交流往往能促进思想的提炼和澄清。然而，现有的假新闻检测模型未能充分融入这种方法，通常将其分析局限于单一、未经挑战的视角。这种局限性阻碍了对新闻内容的全面评估，并可能忽略对准确检测至关重要的细微信号。

为应对这些挑战，我们提出了一种基于大语言模型的新型多智能体系统，命名为TruEDebate（TED）。TED旨在通过遵循正式辩论的结构化方法，提升虚假新闻检测的可解释性和有效性。受“真理在辩论中更为清晰”原则的启发，我们的方法模拟了多个智能体之间动态且结构化的辩论，每个智能体从不同角度审视新闻内容。这种辩论结构遵循林肯-道格拉斯辩论格式，其中智能体通过一系列阶段来陈述、挑战和改进其论点。该框架模拟了人类通过讨论评估信息的认知过程，促进对新闻内容的深度智力参与和综合评估。在此框架中，多个被分配不同角色的智能体，通过结构化辩论过程来评估新闻的真实性。这种辩论过程增强了检测过程的可解释性，并反映了人类辩论的认知过程，即分析并整合相互冲突的观点。

TED框架包含两个创新组件：DebateFlow代理和InsightFlow代理。DebateFlow代理将多个由大语言模型驱动的代理组织成两个对立的团队——一个支持新闻的真实性，另一个反对。这些代理在开启、交叉询问反驳和总结陈词的连续阶段中进行互动，有效地模拟了一个严格的辩论环境。这种设置鼓励探索不同的观点，提炼和澄清思想，并解决了现有模型从单一角度分析信息的局限性。同时，InsightFlow代理包含两个专门的子代理：合成代理和分析代理。合成代理负责总结辩论并提供全面的观点，确保结论的连贯性和完整性。分析代理由一个角色感知编码器和一个辩论图组成，它整合了角色嵌入，并通过交互注意力机制对不同的辩论角色和论点之间的交互进行建模。通过捕捉辩论中固有的详细关系和话语结构，分析代理有效地识别出假新闻检测所需的关键指标。

我们在两个数据集ARG-EN和ARG-CN上进行了全面的实验，以评估TED的有效性。结果表明，我们的框架在多个评估指标上优于传统方法。此外，TED促进了结构化辩论程序，增强了逻辑推理，有助于得出准确的结论。通过实现系统化辩论和深入分析，我们的方法推动了信息系统的发展，这些系统能够做出更精确和透明的决策过程，从而支持创建负责任的信息环境。

我们的贡献可以总结如下：

- 我们介绍了TED，这是一个基于大语言模型的新颖且可解释的假新闻检测框架。TED不仅通过结构化辩论进行检测，还提供可解释的论证，以帮助理解结果。
- 受林肯-道格拉斯辩论理论 [8]，“真理越辩越明”的启发，我们在框架中设计了两个关键组件：合成代理和分析代理。合成代理总结辩论并综合全面观点，确保结论连贯。分析代理结合了角色感知编码器和辩论图，整合角色嵌入，并通过注意力机制模拟不同辩论角色和论点之间的交互。这种结构使分析代理能够识别对准确假新闻检测至关重要的关键指标。

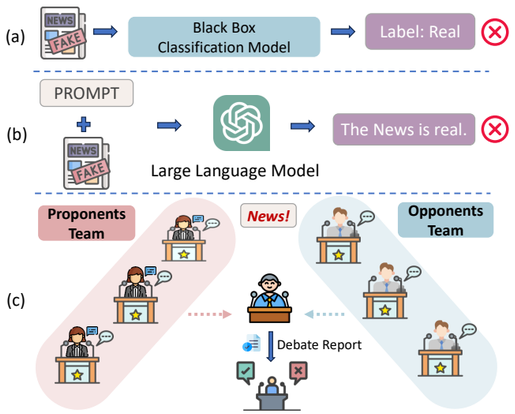


Figure 1: (a) Traditional models primarily rely on black box classification models for detecting fake news. (b) LLMs are prompted to output detection results directly, which do not fully utilize the potential of large model reasoning. (c) Our multi-agents framework simulates a formal debate scenario to engage in discussions on news topics, reaching detection results through debate reasoning and interpretability.

Traditional fake news detection methods predominantly rely on machine learning models, particularly those based on BERT [7], which excel in text analysis [16, 19, 31]. While these models have achieved notable success, they often suffer from low interpretability and limited generalization capabilities, hindering their effectiveness across diverse contexts. Recent approaches have attempted to leverage LLMs by developing specific prompts to generate explanations and detection results [15, 29]. However, these methods may not fully utilize LLMs' inherent reasoning abilities and complex cognitive processes, frequently resulting in excessively simplified and premature conclusions.

In the context of truth-seeking, the proverb "*truth becomes clearer through debate*" emphasizes the importance of examining and challenging information from multiple perspectives to arrive at a more accurate understanding [2, 21]. The exchange of differing viewpoints in a structured debate often leads to refining and clarifying ideas. However, existing fake news detection models fail to incorporate this approach fully, often limiting their analysis to a single, unchallenged perspective. This limitation prevents a comprehensive evaluation of news content and may overlook detailed signals that are critical for accurate detection.

To address these challenges, we introduce a novel multi-agent system with LLMs named TruEDebate (TED). TED is designed to enhance the interpretability and effectiveness of fake news detection by following the structured methodologies of formal debate. Inspired by the principle that truth emerges more clearly through debate, our approach simulates dynamic and structured debates among multiple agents, each examining the news from different angles. This debate structure follows the Lincoln-Douglas Debate format [8], where agents engage in a series of stages to present, challenge, and refine their arguments. This framework mirrors the human cognitive process of evaluating information through discourse, fostering deep intellectual engagement and comprehensive evaluation of news content. In this framework, multiple agents, each assigned distinct roles, engage in a structured debate process to evaluate the truth of news items. This debate process enhances the interpretation of the detection process and reflects the human cognitive process of debate, where conflicting viewpoints are analyzed and integrated.

The TED framework comprises two innovative components: the DebateFlow Agents and the InsightFlow Agents. The DebateFlow Agents organize multiple LLM-powered agents into two opposing teams—one supporting and the other opposing the truth of the news. These agents engage in sequential stages of opening, cross-examination rebuttal, and closing statements, effectively simulating a rigorous debate environment. This setup encourages the exploration of differing viewpoints, refining, and clarifying ideas, and addresses the limitation of existing models that analyze information from a single perspective. Concurrently, the InsightFlow Agents encompass two specialized sub-agents: the Synthesis Agent and the Analysis Agent. The Synthesis Agent is responsible for summarizing the debates and providing a comprehensive viewpoint, ensuring a coherent and thorough conclusion. The Analysis Agent, composed of a role-aware encoder and a debate graph, integrates role embeddings and models the interactions between different debate roles and arguments through an interaction attention mechanism. By capturing the detailed relationships and discourse structures inherent in the debates, the Analysis Agent effectively identifies critical indicators essential for fake news detection.

We conducted comprehensive experiments on two datasets, ARG-EN and ARG-CN, to evaluate the effectiveness of TED. The results indicate that our framework outperforms traditional methods across multiple evaluation metrics. Additionally, TED facilitates structured debate procedures and enhances logical reasoning, contributing to accurate conclusions. By enabling systematic debate and in-depth analysis, our approach advances the development of information systems capable of making more precise and transparent decision-making processes, thereby supporting the creation of responsible information environments.

Our contribution can be summarized in the following ways:

- We introduce TED, a novel and interpretable fake news detection framework based on LLMs. TED not only performs detection through structured debates but also provides interpretable justifications to aid in understanding the results.
- Inspired by Lincoln-Douglas debate theory [8], which holds that “truth becomes clearer through debate," we design two key components within our framework: the Synthesis Agent and the Analysis Agent. The Synthesis Agent summarizes the debates and synthesizes comprehensive viewpoints, ensuring coherent conclusions. The Analysis Agent, incorporating a role-aware encoder and a debate graph, integrates role embeddings and models the interactions between different debate roles and arguments through an attention mechanism. This structure enables the Analysis Agent to identify critical indicators essential for accurate fake news detection.

•在ARG-EN和ARG-CN数据集上的大量实验表明，TED相较于传统方法具有优越的性能，展现了其做出准确和透明的决策的能力。此外，TED在不同的大语言模型（LLM）骨干网络上也保持了稳健的性能，展示了其在假新闻检测中的适应性和泛化能力。

2 相关工作

2.1 假新闻检测

假新闻检测是打击虚假信息的关键步骤 [12]。该领域早期的相关工作包括钱等人的研究 [36], ，该研究专注于假新闻的早期检测，仅考虑检测时可用的新闻文章文本。余等人 [49] 应用卷积方法进行虚假信息识别。随着该领域的发展，人们引入了各种方法来提高检测效果。例如，金等人 [17] 通过更好地反映人类思维的逻辑过程，并能够对细微线索进行建模，转向对假新闻进行细粒度推理。此外，郑等人 [60] 研究了考虑其相互关系复杂性的多模态数据集成。最近，随着大语言模型的兴起，胡等人 [15] 结合了大和小语言模型，仅使用文本内容就实现了对假新闻检测的良好改进。李等人 [25] 设计的大语言模型基础上的智能体工作流使用工具学习和知识检索来检测假新闻。刘等人 [27, 28] 采用多智能体系统来模拟假新闻的传播和演化。

然而，他们并未充分利用大语言模型的潜力。我们的研究通过进一步利用由多智能体组成、配备大语言模型的辩论系统，增强了假新闻检测的能力。在我们的方法中，我们不调用任何外部工具，也不需要基于知识的检索，因此可以轻松地迁移到任何假新闻检测场景中。

2.2 基于LLM的多智能体讨论

尽管大语言模型取得了显著进展，但关于多智能体讨论的研究仍处于早期阶段，旨在利用超越单个模型的集体能力。该领域的一项关键贡献是ChatEval [3], ，它设计了一支多智能体裁判团队，独立地对各种模型在开放式问题和标准自然语言生成任务上生成的响应质量进行商讨和评估。其他研究强调通过多样化智能体角色来丰富辩论过程 [24, 33]。Liang等人 [26] 进一步探讨了角色分配如何影响多智能体交互的有效性。Chen等人 [4] 引入了ReConcile框架，该框架使用不同的LLM和加权投票系统来达成共识，捕捉更广泛的观点。Du等人 [9] 采用多智能体辩论来提升LLM的数学能力。此外，Zhang等人 [52]从社会心理学的角度分析了多智能体讨论，以理解社会因素如何影响智能体之间的协作和推理过程。

然而，这些工作并未真正描绘出其框架内的辩论过程，也无法从多个角度分析问题。在此基础上，我们的工作引入了一个结构化的辩论框架，该框架利用多智能体交互来提升假新闻检测的可解释性和有效性。

2.3 基于大语言模型的多智能体系统

凭借大语言模型强大的文本生成能力，将大语言模型集成到模拟社会动态中代表了一个蓬勃发展的研究领域，并取得了有前景的成果 [18, 23, 39, 45]。这些基于大语言模型的生成式智能体在数字环境中表现出色，在自然语言任务 [5, 6, 50] 和推理任务 [51]中展现出专业能力。Törnberg 等人 [42]使用大语言模型和基于智能体的建模来模拟社交媒体，重点关注新闻推送算法，并提供了现实世界的见解。此外，Park 等人 [34] 证明了基于大语言模型的智能体可以生成与人类生产的内容无法区分的社交媒体内容。此外，Qian 等人 [35]设计了ChatDev，一个模拟人类开发团队的虚拟开发公司。

这些利用大语言模型多智能体系统解决复杂任务的框架都共享一个关键组件：多个智能体之间的交互 [3, 14, 30, 41, 46, 54–56]。然而，这些研究只考虑了不同智能体之间的消息传递。虽然一些研究声称进行多智能体辩论，但它们并未充分描绘交互过程中的辩论过程。我们的工作建立了一个标准辩论场景，其中不同角色从不同角度辩论新闻的真实性，以确定新闻主题的真伪。据我们所知，这是在虚假新闻背景下首次进行社会辩论模拟的研究。

3 方法论

3.1 任务形式化

我们将虚假新闻检测重新形式化为多个大语言模型智能体之间的交互式辩论。我们引入一个由 N 个大语言模型智能体组成的系统，记为 $\mathrm{A} = (a_1, \ldots, a_N)$，每个智能体在辩论过程中被分配独特的角色和立场。给定一个新闻项 F，所有智能体将进行结构化的辩论以确定其真实性。这些智能体被组织成两个对立的团队：

- **支持者**：认为 F 为真的代理。
- **反对者**：认为 F 为虚假的代理。

每个代理在辩论中扮演特定角色，并参与预定义的阶段。所有交互都会记录在辩论日志 D中。在辩论的结论阶段，有两种类型的代理评估结果。合成代理分析辩论日志 D 以生成综合摘要报告 S。同时，分析代理利用摘要报告 S 和原始新闻项 F来预测真实性值。

我们的目标是在这一辩论框架中利用大语言模型的推理能力，以提高假新闻检测的有效性和可解释性。

• Extensive experiments on the ARG-EN and ARG-CN datasets demonstrate TED's superior performance to traditional methods, showing its ability to make accurate and transparent decisions. Furthermore, TED also maintains robust performance across different LLM backbones, showcasing its adaptability and generalizability in fake news detection.

2 Related Work

2.1 Fake News Detection

Fake news detection is an essential step in the fight against misinformation [12]. Earlier works in this area include the study by Qian et al. [36], which focuses on the early detection of fake news, considering only the text of news articles available at the time of detection. Yu et al. [49] applied a convolutional approach for misinformation identification. As the field evolved, various methods have been introduced to improve detection efficacy. For example, Jin et al. [17] move towards fine-grained reasoning for fake news detection by better reflecting the logical processes of human thinking and enabling the modeling of subtle clues. Furthermore, Zheng et al. [60] investigated the integration of diverse multi-modal data considering the complexity of their interrelations. Recently, with the rise of LLMs, Hu et al. [15] combined large and small LMs and achieved good improvements using only textual content for fake news detection. The agent workflow based on a large language model designed by Li et al. [25] uses tool learning and knowledge retrieval to detect fake news. Liu et al. [27, 28] employs a multi-agent system to simulate the propagation and evolution of fake news.

However, they did not fully utilize the potential of LLMs. Our research enhances the capability of fake news detection by further leveraging a debate system composed of multiple agents with LLMs. In our approach, we do not call any external tools, nor do we require knowledge-based retrieval, making it easily transferable to any fake news detection scenario.

2.2 Multi-agent Discussion with LLMs

Despite significant advancements in LLMs, research on multi-agent discussions, where multiple LLMs interactively collaborate to solve problems, is still in its early stages, aiming to leverage collective capabilities beyond individual models. A key contribution in this field is ChatEval [3], which designs a team of multi-agent referees to independently deliberate and assess the quality of responses generated by various models on open-ended questions and standard natural language generation tasks. Other studies emphasize diversifying agent roles to enrich the debate process [24, 33]. Liang et al. [26] further explores how role assignment impacts the effectiveness of multi-agent interactions. Chen et al. [4] introduce the ReConcile framework, which uses different LLMs and a weighted voting system to reach consensus, capturing a broader range of perspectives. Du et al. [9] employs multi-agent debate to improve the mathematical abilities of LLMs. Additionally, Zhang et al. [52] analyzes multi-agent discussions from a social psychology perspective to understand how social factors influence collaboration and reasoning processes among agents.

However, these works do not truly depict a debate process within their frameworks and cannot analyze problems from multiple perspectives. Building upon these foundations, our work introduces a structured debate framework that leverages multi-agent interactions to enhance the interpretability and effectiveness of fake news detection.

2.3 LLM-Based Multi-Agent System

With LLMs' powerful text generation capabilities, integrating LLMs into simulating social dynamics represents a burgeoning field of research, yielding promising results [18, 23, 39, 45]. These LLM-based generative agents excel in digital environments, demonstrating proficiency in natural language tasks [5, 6, 50] and reasoning task [51]. Törnberg et al. [42] used LLMs and agent-based modeling to simulate social media, focusing on news feed algorithms and offering real-world insights. Further, Park et al. [34] demonstrated that LLM-based agents could generate social media content indistinguishable from that produced by humans. Moreover, Qian et al. [35] designed ChatDev, a virtual development company that simulates a human development team.

These frameworks that utilize LLM multi-agent systems to solve complex tasks all share a crucial component: the interaction among multiple agents [3, 14, 30, 41, 46, 54–56]. However, these works consider only the message passing between different agents. While some claim to engage in multi-agent debates, they do not adequately depict the debate process during interactions. Our work establishes a standard debate scenario, where various roles debate fake news from different perspectives to determine the truth of the news topic. To the best of our knowledge, this is the first instance of a social debate simulation in the context of fake news.

3 Methodology

3.1 Task Formulation

We reformulate fake news detection as an interactive debate among multiple LLM agents. We introduce a system comprising N LLM agents denoted as $\mathcal{A} = (a_1, \ldots, a_N)$, each assigned unique roles and stances in the debate process. Given a news item F, all agents engage in a structured debate to determine its veracity. The agents are organized into two opposing teams:

- **Proponents**: Agents who argue that F is true.
- **Opponents**: Agents who argue that F is fake.

Each agent plays a specific role in the debate and participates in predefined stages. All interactions are recorded in a debate log D. At the conclusion stage of the debate, two types of agents evaluate the outcome. The Synthesis Agent analyzes the debate log D to produce a comprehensive summary report S. At the same time, the Analysis Agent utilizes the summary report S and the original news item F to predict the truth value.

Our objective is to leverage LLMs' reasoning capabilities within this debate framework to enhance the effectiveness and interpretability of fake news detection.

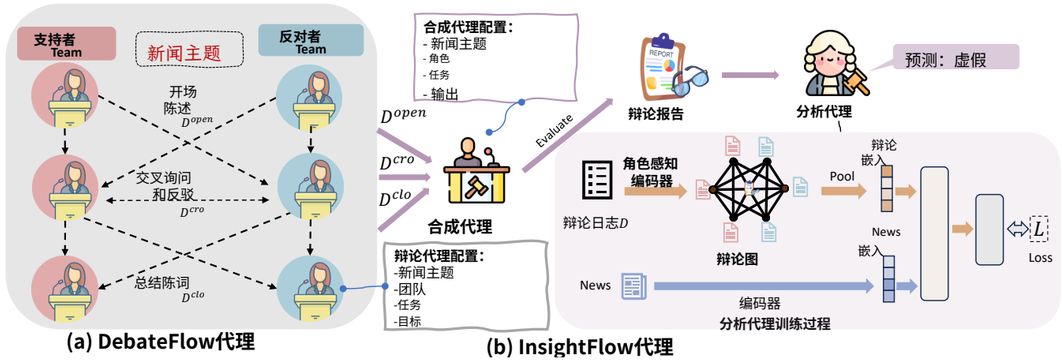


图2：我们的TED框架由两个主要组件组成：(a) DebateFlow代理和(b) InsightFlow代理。DebateFlow代理模拟辩论场景中的推理过程，包括开场、交叉询问、反驳和总结陈词。InsightFlow代理由合成代理和分析代理组成。前者根据所有辩论记录生成辩论报告，然后将其传递给后者以预测新内容是否为真。

3.2 我们的Multi-Agents辩论框架

我们介绍了**TruEDebate (TED)**，这是一个通过模拟多个由大语言模型驱动的智能体之间进行结构化辩论来增强假新闻检测的框架。TED由两个主要组件组成：

•**DebateFlow代理**: 分为两个对立团队——支持者，他们认为新闻项为真，以及反对者，他们认为其为虚假。受林肯-道格拉斯辩论理论 [8], 启发，这些代理参与多阶段的辩论过程，包括开场、交叉询问、反驳和总结陈词。这种设置模拟了正式辩论场景，鼓励从多个视角对新闻进行彻底审查。

• **InsightFlow代理**: 由 合成代理 和 分析代理 组成。合成代理总结辩论，提供对关键论点和反论点的连贯概述。分析代理采用角色感知编码器和辩论图来模拟辩论中的交互和话语结构，最终预测新闻的真实性。

如图2所示，通过这些智能体的交互，TED促进了深度的智力参与和逻辑推理，从而实现更准确且可解释的假新闻检测。结构化的辩论过程不仅提升了智能体对新闻内容的理解，也为最终决策提供了透明的理由，体现了“真理越辩越明”的原则。

3.3 DebateFlow代理

DebateFlow代理是TED框架的核心，负责组织结构化的辩论以评估新闻的真实性 F。代理被分为两个对立的团队：支持者（T）（他们认为 F是真的），以及反对者（Topp）（他们认为 F是虚假的）。每个团队都由在辩论不同阶段担任特定角色的代理组成。

受林肯-道格拉斯辩论 [8]，（通常用于辩论环境中解决有争议问题的辩论形式）的启发，辩论被分为三个主要阶段：开场、交叉询问和反驳、以及结束。令 $\Phi = \{\phi_1, \phi_2, \phi_3\}$表示这些阶段。在每个阶段 $\phi_k \in \Phi$，来自两个团队的代理根据其角色参与。在开场阶段（ϕ_1），代理们提出支持其团队立场的初步论点。每个代理 a_i 基于新闻项 F和其团队的立场生成一个论点 $d_{(1)\,i}$：

$$d_i^{(1)} = f_{arg}(F, Stance_i), \tag{1}$$

其中 f_{arg} 代表由大语言模型实现的说服性提示函数，而 $Stance_i \in \{True,\ \ Fake\}$则对应团队的立场。

在交叉询问与反驳阶段(ϕ_2)，代理通过提问和反驳来挑战对方团队的论点。来自一个团队的代理a_j 针对对方团队在上一阶段提出的论点，制定一个提问或反驳$d_{(2)\ j}$：

$$d_j^{(2)} = f_{reb}(D^{(1)}, Stance_j), \tag{2}$$

其中 f_{reb} 是交叉询问与反驳提示函数，而 $D_{(1)}$ 是阶段 ϕ_1中的论点集合。

在 总结阶段(ϕ_3)，代理总结其团队的立场并强化关键要点。每个代理 a_k 产出一个总结性陈述 $d_{(3)\,k}$：

$$d_k^{(3)} = f_{clo}(D^{(1)} \cup D^{(2)}, Stance_k), \tag{3}$$

其中 f_{clo}是总结提示函数，而 $D_{(1)} \cup D_{(2)}$代表从上一阶段累积的辩论内容。

辩论日志 D 记录了辩论过程中的所有交互：

$$D = \bigcup_{k=1}^{3} D^{(k)}, \quad \text{with} \quad D^{(k)} = \{d_i^{(k)} \mid a_i \in \mathcal{A}^{(k)}\}, \tag{4}$$

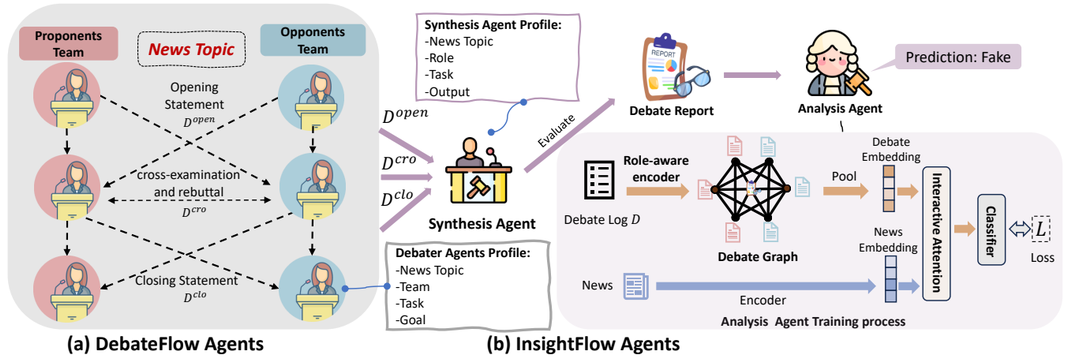


Figure 2: Our TED framework consists of two main components: (a) the DebateFlow Agents and (b) the InsightFlow Agents. The DebateFlow Agents simulate the reasoning process in a debate scenario, including the Opening, Cross-examination, Rebuttal, and Closing Statement. The InsightFlow Agents consist of the Synthesis Agent and the Analysis Agent. The former generates a debate report based on all debate records, which is then passed to the latter to predict whether the new content is true.

3.2 Our Multi-Agents Debate Framework

We introduce **TruEDebate (TED)**, a framework that enhances fake news detection by simulating a structured debate among multiple LLM-powered agents. TED consists of two main components:

• **DebateFlow Agents**: Organized into two opposing teams—the *Proponents*, who argue that the news item is true, and the *Opponents*, who argue that it is fake. Inspired by Lincoln-Douglas debate theory [8], these agents engage in a multi-stage debate process, including opening, cross-examination, rebuttal, and closing statements. This setup mirrors formal debate settings, encouraging thorough examination of the news from multiple perspectives.

• **InsightFlow Agents**: Comprising the *Synthesis Agent* and the *Analysis Agent*. The Synthesis Agent summarizes the debate, providing a cohesive overview of the key arguments and counterarguments. The Analysis Agent employs a role-aware encoder and a debate graph to model the interactions and discourse structures within the debate, ultimately predicting the authenticity of the news item.

As shown in Figure 2, through the interactions of these agents, TED fosters deep intellectual engagement and logical reasoning, leading to more accurate and interpretable fake news detection. The structured debate process not only refines the agents' understanding of the news item but also provides transparent justifications for the final decision, embodying the principle that "truth becomes clearer through debate."

3.3 DebateFlow Agents

The **DebateFlow Agents** form the core of the TED framework, orchestrating a structured debate to assess the authenticity of a news item F. The agents are divided into two opposing teams: the *Proponents* ($\mathcal{T}_{pro}$), who argue that F is true, and the *Opponents* ($\mathcal{T}_{opp}$), who contend that F is fake. Each team consists of agents assigned specific roles corresponding to different debate stages.

Inspired by Lincoln-Douglas debate [8], which is commonly used in debate settings to resolve controversial issues, the debate is structured into three main stages: *Opening*, *Cross-examination and Rebuttal*, and *Closing*. Let $\Phi = \{\phi_1, \phi_2, \phi_3\}$ denote these stages. In each stage $\phi_k \in \Phi$, agents from both teams participate according to their roles. In the *Opening* stage (ϕ_1), agents present their initial arguments supporting their team's stance. Each agent a_i generates an argument $d_i^{(1)}$ based on the news item F and their team's position:

$$d_i^{(1)} = f_{arg}(F, Stance_i), \tag{1}$$

where f_{arg} represents the argumentation prompt function implemented by the LLM, and $Stance_i \in \{True, Fake\}$ corresponds to the team's stance.

In the *Cross-examination and Rebuttal* stage (ϕ_2), agents challenge the opposing team's arguments through questions and rebuttals. An agent a_j from one team formulates a question or rebuttal $d_j^{(2)}$ targeting the arguments made in the previous stage by the opposing team:

$$d_j^{(2)} = f_{reb}(D^{(1)}, Stance_j), \tag{2}$$

where f_{reb} is the Cross-examination and Rebuttal prompt function, and $D^{(1)}$ is the set of arguments in stage ϕ_1.

In the *Closing* stage (ϕ_3), agents summarize their team's position and reinforce key points. Each agent a_k produces a concluding statement $d_k^{(3)}$:

$$d_k^{(3)} = f_{clo}(D^{(1)} \cup D^{(2)}, Stance_k), \tag{3}$$

where f_{clo} is the closing prompt function, and $D^{(1)} \cup D^{(2)}$ represents the accumulated debate content from previous stages.

The debate log D accumulates all interactions during the debate:

$$D = \bigcup_{k=1}^{3} D^{(k)}, \quad \text{with} \quad D^{(k)} = \{d_i^{(k)} \mid a_i \in \mathcal{A}^{(k)}\}, \tag{4}$$

1: **输入：** 新闻项 $F$2: **输出：** 新闻项 F 的预测真实性值 $\hat{y}$，解释原因 $R$3: **初始化DebateFlow代理：** 4: 将代理分配到两个团队：

- 支持者：主张 F 为真
- 反对者：论证 F 为假

5: 在每个团队内分配角色（例如，开场发言人、提问者、总结发言人） 6: DebateFlow代理： 7: 对每个辩论阶段 ϕ 在 [开场陈述、交叉询问、反驳和总结陈词]中执行 8: Agent i 根据角色提出论点或反驳 9: 在辩论日志 D_ϕ中记录交互 10: 结束循环 11: InsightFlow代理： 12: 合成代理: 13: 分析辩论日志 $D = \{D_\phi\}$以生成评估报告 S 14: 分析代理: 15: 对每个辩论交互（节点）n_i 在 D 中执行 16: 使用角色感知编码器获取节点 n_i 嵌入 h_i 17: 结束循环 18: 构建辩论图 $G = (V, E)$，节点 $V = \{h_i\}$ 19: 对 G 应用GATs以获得图嵌入 g 20: 对新闻项 F 进行编码以获得内容嵌入 e_F 21: 组合表示 $c \leftarrow \text{InteractiveAttention}(g, e_F)$ 22: 使用 c预测真实性值 $\hat{y}$ 23: 返回预测的真实性值 $\hat{y}$，解释性理由$R = \{S, D\}$

算法 1: TruEDebate (TED)：用于假新闻检测的多代理辩论框架

其中 $A^{(k)}$是参与第 ϕ_k 阶段的代理集合。

每个代理的贡献 $d_i^{(k)}$取决于新闻项 F、其团队的立场、其在辩论中的角色以及先前的辩论上下文$D^{<k}$：

$$d_i^{(k)} = f_{\text{LLM}}(F, D^{<k}, \text{Role}_i, \text{Stance}_i), \tag{5}$$

其中 $D^{<k} = \bigcup_{m=1}^{k-1} D^{(m)}$ 表示截至第 ϕ_{k-1} 阶段的辩论历史，Role_i 指定了代理的角色。

整个算法过程可以用算法 1 描述。这种结构化交互确保代理既考虑新闻内容，也考虑演变的辩论上下文，从而促进推理的一致性和深度。通过参与这一过程，DebateFlow代理模拟了对新闻项 F 的严格审查，反映了人类辩论的动态，并从多个视角促进了综合评估。

所有功能提示都可以在我们的代码文件中找到。

3.4 InsightFlow代理

DebateFlow代理进行辩论后，**Insight-Flow代理**分析累积的讨论内容，对新闻项的真实性做出最终预测 F。该组件由两个主要代理构成：**合成代理**和**分析代理**。

3.4.1 **合成代理。**合成代理处理辩论日志 D（包含所有辩论阶段的互动内容），生成一份简洁且全面的总结 S，涵盖双方提出的主要论点和反驳论点。该总结从假新闻检测的多个视角进行评估，以确保判断的全面性。形式上，合成代理的核心功能提示定义如下：

它应该包含对辩论的详细评估说明。重点关注通过以下方面评估所涉及新闻的真实性：1. 新闻是否包含具体细节和可验证信息。2. 新闻是否引用了可靠来源或新闻机构。3. 新闻的语气和风格，真实新闻通常更客观中立。4. 是否使用情绪化语言，这可能是假新闻的特征。5. 新闻中的信息是否可以通过其他可靠渠道得到确认。

合成代理的目的是总结辩论的关键点，提供一份连贯的概述，突出每方论点的优缺点。这份总结 S 连接了辩论话语和后续的分析过程，确保最相关信息易于获取，以便进一步评估。

3.4.2 **分析代理.** 分析代理将辩论互动与原始新闻内容相结合，以预测新闻项目 F的真实性值$\hat{y}$。该代理采用角色感知编码器、辩论图和新闻-辩论交互注意力机制的组合，以捕捉语义内容、辩论关系动态及其与新闻内容的相关性。

角色感知编码器。 为了捕捉不同辩论角色的独特特征，我们设计了一个角色感知编码器。对于每个辩论交互 d_i，节点表示是通过将编码器的 [CLS]token 嵌入与一个投影的角色嵌入连接构建的：

$$\begin{aligned} \mathbf{h}_i^{\text{enc}} &= \text{Encoder}(\text{Text}_i), \\ \mathbf{r}_i^{\text{proj}} &= \mathbf{W}_{\text{role}}\, \mathbf{e}_i, \\ \mathbf{h}_i^{\text{node}} &= \left[\mathbf{h}_i^{\text{enc}} ; \mathbf{r}_i^{\text{proj}} \right], \end{aligned} \tag{6}$$

其中 $\mathbf{h}\text{enc}_i \in \mathbb{R}^{d_h}$ 是来自编码器的 [CLS]token 表示，$\mathbf{e}_i \in \mathbb{R}^{d_r}$ 是角色的嵌入 r_i，$\mathbf{W}\text{role} \in \mathbb{R}^{d_h \times d_r}$ 将角色嵌入投影到匹配编码器的维度，以及 [;] 表示向量连接。得到的节点表示 $\mathbf{h}\text{node}_i$ 集成了每个辩论交互的语义内容和角色特定信息。

辩论图. 编码后的节点表示被结构化为一个辩论图 $G = (V, E)$, 其中节点 $V = \{h\text{节点}_i\}$

Input: News item F
Output: Predicted truth value $\hat{y}$ of news item F, Explanatory reasons R
Initialize DebateFlow Agents:
Assign agents into two teams:

- *Proponents*: Support that F is true
- *Opponents*: Argue that F is fake

Assign roles within each team (e.g., Opening Speaker, Questioner, Closing Speaker)
DebateFlow Agents:
for each debate stage ϕ in [Opening Statement, Cross-examination, Rebuttal, and Closing Statement] **do**
  Agent i presents arguments or rebuttals based on the role
  Record interactions in debate log D_ϕ
end for
InsightFlow Agents:
Synthesis Agent:
Analyze debate logs $D = \{D_\phi\}$ to produce evaluation report S
Analysis Agent:
for each debate interaction (node) n_i in D **do**
  Obtain node n_i embedding h_i with a Role-aware Encoder
end for
Build Dabate graph $G = (V, E)$ with nodes $V = \{h_i\}$
Apply GATs to G to obtain graph embedding g
Encode news item F to obtain content embedding e_F
Combined representation $c \leftarrow \text{InteractiveAttention}(g, e_F)$
Predict truth value $\hat{y}$ using c
return Predicted truth value $\hat{y}$, explanatory reasons $R = \{S, D\}$

Algorithm 1: TruEDebate (TED): Multi-Agent Debate Framework for Fake News Detection

where $\mathcal{A}^{(k)}$ is the set of agents participating in stage ϕ_k.

Each agent's contribution $d_i^{(k)}$ depends on the news item F, their team's stance, their role in the debate, and the prior debate context $D^{<k}$:

$$d_i^{(k)} = f_{\text{LLM}}(F, D^{<k}, \text{Role}_i, \text{Stance}_i), \tag{5}$$

where $D^{<k} = \bigcup_{m=1}^{k-1} D^{(m)}$ represents the debate history up to stage ϕ_{k-1}, and Role_i specifies the agent's role.

The entire algorithm process can be described in Algorithm 1. This structured interaction ensures that agents consider both the news content and the evolving debate context, promoting coherence and depth in their reasoning. By engaging in this process, the DebateFlow Agents simulate a rigorous examination of the news item F, mirroring the dynamics of human debates and fostering a comprehensive evaluation from multiple perspectives.

The whole set of function prompts can be found in our code files.

3.4 InsightFlow Agents

After the debate conducted by DebateFlow Agents, the **Insight-Flow Agents** analyze the accumulated discourse to produce a final prediction on the authenticity of the news item F. This component consists of two principal agents: the **Synthesis Agent** and the **Analysis Agent**.

3.4.1 ***Synthesis Agent***. The Synthesis Agent processes the debate log D, which contains all interactions from the debate stages, to generate a concise and comprehensive summary S of the key arguments and counterarguments presented by both teams. This summary is evaluated from multiple perspectives of fake news detection to ensure a well-rounded judgment. Formally, the core function prompt of the Synthesis Agent is defined as follows:

It should contain a detailed explanation of your assessment of the debate. Focus on evaluating the authenticity of the news involved in the topic by checking the following:
1. Whether the news contains specific details and verifiable information.
2. Whether the news cites reliable sources or news organizations.
3. The tone and style of the news, with real news generally being more objective and neutral.
4. Any use of emotional language, which might be a characteristic of fake news.
5. Whether the information in the news can be confirmed through other reliable channels.

The purpose of the Synthesis Agent is to encapsulate the critical points of the debate, providing a coherent overview highlighting the strengths and weaknesses of each side's arguments. This summary S bridges the debate discourse and the subsequent analytical processes, ensuring that the most pertinent information is readily accessible for further evaluation.

3.4.2 ***Analysis Agent***. The Analysis Agent integrates the debate interactions with the original news content to predict the truth value $\hat{y}$ of the news item F. This agent employs a combination of role-aware encoder, debate graph, and a news-debate interactive attention mechanism to capture both semantic content and the relational dynamics of the debate, as well as their relevance to the news content.

Role-aware Encoder. To capture the distinct characteristics of different debate roles, we design a role-aware encoder. For each debate interaction d_i, the node representation is constructed by concatenating the encoder's `[CLS]` token embedding with a projected role embedding:

$$\begin{aligned} \mathbf{h}_i^{\text{enc}} &= \text{Encoder}(\text{Text}_i), \\ \mathbf{r}_i^{\text{proj}} &= \mathbf{W}_{\text{role}}\, \mathbf{e}_i, \\ \mathbf{h}_i^{\text{node}} &= \left[\mathbf{h}_i^{\text{enc}} ; \mathbf{r}_i^{\text{proj}} \right], \end{aligned} \tag{6}$$

where $\mathbf{h}_i^{\text{enc}} \in \mathbb{R}^{d_h}$ is the `[CLS]` token representation from the encoder, $\mathbf{e}_i \in \mathbb{R}^{d_r}$ is the embedding for role r_i, $\mathbf{W}_{\text{role}} \in \mathbb{R}^{d_h \times d_r}$ projects the role embedding to match the encoder's dimension, and [;] denotes vector concatenation. The resulting node representation $\mathbf{h}_i^{\text{node}}$ integrates both semantic content and role-specific information of each debate interaction.

Debate Graph. The encoded node representations are structured into a debate graph $G = (V, E)$, where the nodes $V = \{h_i^{\text{node}}\}$

代表辩论交互，而边 E 则代表交互之间的关系，例如顺序顺序或明确引用。

我们采用图注意力网络（GAT）[43] 处理节点表示。GAT层通过关注其邻居来更新每个节点的表示：

$$\mathbf{h}_i^{(l+1)} = \sigma\left(\sum_{j\in\mathcal{N}(i)} \alpha_{ij}^{(l)}\mathbf{W}^{(l)}\mathbf{h}_j^{(l)}\right), \tag{7}$$

其中 $\mathbf{h}(l)i$ 是节点 i 在层 l的表示，N (i) 表示节点 i的邻居集合，$\mathbf{W}(l)$ 是一个可学习的权重矩阵,σ是一个激活函数，而 $\alpha(l)ij$是注意力系数。通过堆叠多个GAT层，模型捕获了高阶交互和辩论中的关系动态。

辩论-新闻交互注意力。 为了将辩论表示与新闻内容集成，我们采用了一种交互式注意力机制。首先，我们编码新闻内容 F 并应用全局池化以获得图级辩论表示$\mathbf{g}$：

$$\begin{aligned}\mathbf{e}_F &= \text{Encoder}(F),\\ \mathbf{g} &= \text{GlobalPool}(\mathbf{H}^{(L)}),\end{aligned} \tag{8}$$

其中 $\mathbf{H}(L)$ 是第 L-个GAT层的输出，GlobalPool 表示一个全局平均池化操作。

接下来，我们将辩论表示和新闻内容嵌入投影到共同维度：

$$\begin{aligned}\mathbf{g}^{\text{proj}} &= \mathbf{W}_g\mathbf{g},\\ \mathbf{e}_F^{\text{proj}} &= \mathbf{W}_e\mathbf{e}_F,\end{aligned} \tag{9}$$

其中 $\mathbf{W}g$ 和$\mathbf{W}e$ 是投影矩阵。

然后我们使用多头注意力（MHA）计算交互表示 $\mathbf{c}$ ，以捕捉辩论和新闻内容之间的交互：

$$\mathbf{c} = \text{MHA}(\mathbf{e}_F^{\text{proj}}, \mathbf{g}^{\text{proj}}, \mathbf{g}^{\text{proj}}), \tag{10}$$

最后，我们将投影的辩论表示与交互表示连接起来，以获得综合表示：

$$\mathbf{h} = [\mathbf{g}^{\text{proj}}; \mathbf{c}]. \tag{11}$$

这种交互式注意力允许模型关注与新闻内容相关的辩论的不同方面，有效地整合来自两个来源的信息。

代理训练。 组合表示$\mathbf{h}$ 通过一个全连接层后接softmax激活处理，以生成对类别的预测概率分布。模型通过最小化预测概率与实际标签之间的交叉熵损失进行训练：

$$\begin{aligned}\hat{\mathbf{y}} &= \text{softmax}(\mathbf{W}_{\text{fc}}\mathbf{h} + \mathbf{b}_{\text{fc}}),\\ \mathcal{L} &= -\sum_k y_k \log \hat{y}_k,\end{aligned} \tag{12}$$

where $\mathbf{W}$fc 和 $\mathbf{b}$fc 分别表示分类层的权重和偏置，$\hat{}\mathbf{y}$ 表示预测的概率分布， y_k 是类别 k 的真实标签，而 $\hat{}y_k$ 是类别 k 的预测概率。

4 实验

4.1 实验设置

4.1.1 数据集。 为了评估，我们使用了两个数据集：ARG-EN 和 ARG-CN [15]。这些数据集源自中文数据集微博21 [32]和英文数据集 GossipCop [38]。在之前的工作 [15], 中，他们通过去重和按时间划分数据来处理这些数据集。这种方法有助于防止由于潜在的数据泄露而高估小语言模型（SLM）的性能。我们在实验中使用了这两个数据集。数据集的统计细节汇总在表 2 中。

4.1.2 基线。 遵循先前工作 [15], , 我们进行了比较三种类型方法的实验。

• **仅大语言模型方法**：直接使用 GPT-3.5-turbo 和 GPT-4o- mini 作为大语言模型进行假新闻检测。

• **仅小语言模型方法**：小语言模型（SLMs），如基于 BERT 的模型，在假新闻检测任务上通常表现良好。在此类别中，我们包括以下方法：

(1) Bert [7]: 使用微调的 vanilla BERT-base 模型进行假新闻检测。

(2) EANN [44]: 该模型采用辅助对抗训练，有效隔离并减少事件相关特征，使用出版年份作为辅助任务的标签。

(3) Publisher-Emo [53]: 一种整合多种情感属性与文本特征的模型，用于检测假新闻。

(4)ENDEFA [61]: 该模型采用因果学习消除实体偏差，增强其在分布偏移的假新闻数据集上的泛化能力。该类别中所有方法均使用相同的SLM（BERT）进行文本编码。

• **LLM+SLM方法**。LLM单独使用时往往在该任务上表现不佳。因此，许多研究将LLM与SLM结合以提高性能。

(1) Bert+原理: 它结合新闻和原理编码器的特征，将其输入到MLP中进行预测。

(2) SuperICL [47]: 它将SLM作为插件使用，通过将每个测试样本的预测和置信度水平整合到提示中，增强LLM的上下文学习能力。

(3) ARG [15]: 该模型设计了一个用于假新闻检测的自适应理由引导网络，其中小语言模型从大语言模型提供的理由中，有选择地提取新闻分析的见解。

(4) ARG-D [15]: 它是通过蒸馏创建的ARG的无理由版本，专为不需要查询大语言模型的成本敏感场景而设计。

4.1.3 实现细节。 我们使用Python来运行我们的TED框架。所使用的大语言模型是GPT-4o-mini，通过OpenAI API调用获取，因为它具有代表性且调用方便。代理及其所处的场景使用Python库Mesa [20]定义。对于分析代理，我们利用一个微调的BERT模型，因为近期研究表明它在该领域有广泛应用 [19]。在我们的实验中，我们使用Transformers包中的模型进行中文评估*chinese-bert-wwm-ext* ，以及英文评估*bert-base-uncased* 。优化使用Adam [22], 配合全面的网格搜索进行，

represent the debate interactions, and the edges E represent the relationships between interactions, such as sequential order or explicit references.

We employ Graph Attention Networks (GAT) [43] to process the node representations. The GAT layers update each node's representation by attending to its neighbors:

$$\mathbf{h}_i^{(l+1)} = \sigma\left(\sum_{j\in\mathcal{N}(i)} \alpha_{ij}^{(l)}\mathbf{W}^{(l)}\mathbf{h}_j^{(l)}\right), \tag{7}$$

where $\mathbf{h}_i^{(l)}$ is the representation of node i at layer l, $\mathcal{N}(i)$ denotes the set of neighbors of node i, $\mathbf{W}^{(l)}$ is a learnable weight matrix, σ is an activation function, and $\alpha_{ij}^{(l)}$ are the attention coefficients. By stacking multiple GAT layers, the model captures higher-order interactions and the relational dynamics within the debate.

Debate-News Interactive Attention. To integrate the debate representations with the news content, we employ an interactive attention mechanism. First, we encode the news content F and apply global pooling to obtain the graph-level debate representation $\mathbf{g}$:

$$\begin{aligned}\mathbf{e}_F &= \text{Encoder}(F),\\ \mathbf{g} &= \text{GlobalPool}(\mathbf{H}^{(L)}),\end{aligned} \tag{8}$$

where $\mathbf{H}^{(L)}$ is the output of the L-th GAT layer, and `GlobalPool` denotes a global mean pooling operation.

Next, we project both the debate representation and the news content embedding to a common dimensionality:

$$\begin{aligned}\mathbf{g}^{\text{proj}} &= \mathbf{W}_g\mathbf{g},\\ \mathbf{e}_F^{\text{proj}} &= \mathbf{W}_e\mathbf{e}_F,\end{aligned} \tag{9}$$

where $\mathbf{W}_g$ and $\mathbf{W}_e$ are projection matrices.

We then compute the interaction representation $\mathbf{c}$ using multi-head attention (MHA) to capture the interactions between the debate and news content:

$$\mathbf{c} = \text{MHA}(\mathbf{e}_F^{\text{proj}}, \mathbf{g}^{\text{proj}}, \mathbf{g}^{\text{proj}}), \tag{10}$$

Finally, we concatenate the projected debate representation with the interaction representation to obtain the comprehensive representation:

$$\mathbf{h} = [\mathbf{g}^{\text{proj}}; \mathbf{c}]. \tag{11}$$

This interactive attention allows the model to focus on different aspects of the debate relevant to the news content, effectively integrating information from both sources.

Agent Training. The combined representation $\mathbf{h}$ is processed through a fully connected layer followed by a softmax activation to generate the predicted probability distribution over classes. The model is trained by minimizing the cross-entropy loss between the predicted probabilities and the actual labels:

$$\begin{aligned}\hat{\mathbf{y}} &= \text{softmax}(\mathbf{W}_{\text{fc}}\mathbf{h} + \mathbf{b}_{\text{fc}}),\\ \mathcal{L} &= -\sum_k y_k \log \hat{y}_k,\end{aligned} \tag{12}$$

where $\mathbf{W}_{\text{fc}}$ and $\mathbf{b}_{\text{fc}}$ denote the weights and bias of the classification layer, $\hat{\mathbf{y}}$ represents the predicted probability distribution, y_k is the ground truth label for class k, and $\hat{y}_k$ is the predicted probability for class k.

4 Experiment

4.1 Experiment Setup

4.1.1 Datasets. For evaluation, we utilized two datasets: ARG-EN and ARG-CN [15]. These datasets are derived from the Chinese dataset Weibo21 [32] and the English dataset GossipCop [38]. In previous work [15], they processed these datasets by deduplication and dividing the data temporally. This approach helps prevent overestimating the SLM performance due to potential data leakage. We used these two datasets in the experiment. The statistical details of the datasets are summarized in Table 2.

4.1.2 Baselines. Following previous work [15], we conducted experiments comparing three types of methods.

• **LLM-Only Methods**: Directly use GPT-3.5-turbo and GPT-4o-mini as a LLM for fake news detection.

• **SLM-Only Methods**: Small Language Models (SLMs), like BERT-based models, generally perform well on fake news detection tasks. In this category, we include the following approaches:

(1) Bert [7]: Detecting fake news using a fine-tuned vanilla BERT-base model.

(2) EANN [44]: This model employs auxiliary adversarial training to effectively isolate and reduce event-related features, using the publication year as the label for the auxiliary task.

(3) Publisher-Emo [53]: A model that integrates a range of emotional attributes with textual features to detect fake news.

(4) ENDEFA [61]: A model that employs causal learning to eliminate entity bias, enhancing its generalization capabilities on fake news datasets with distribution shifts. All approaches in this category utilize the same SLM (BERT) for text encoding.

• **LLM+SLM Methods**. LLM solely often does not perform well on the task. Hence, a number of works combine LLM with SLM to enhance performance.

(1) Bert+Rationale: It combines features from both the news and the rationale encoder, feeding them into an MLP for prediction.

(2) SuperICL [47]: It utilizes the SLM as a plug-in to enhance the in-context learning capabilities of the LLM by incorporating predictions and confidence levels for each test sample into the prompt.

(3) ARG [15]: This model designs an Adaptive Rationale Guidance network for fake news detection, where SLMs selectively derive insights on news analysis from the rationales provided by LLMs.

(4) ARG-D [15]: It is a rationale-free version of ARG created through distillation, designed for cost-sensitive scenarios that do not require querying LLMs.

4.1.3 Implementation Details. We use Python to operationalize our framework TED. The LLM used is `GPT-4o-mini` accessed via OpenAI API calls due to its representativeness and convenient calling. Agents and the scene they live in were defined using a Python library Mesa [20]. For Analysis Agent, We utilize a fine-tuned BERT model due to its extensive application in this domain as highlighted by recent studies [19]. For our experiments, we employ models from the Transformers package *chinese-bert-wwm-ext* for Chinese and *bert-base-uncased* for English evaluations. Optimization is performed using Adam [22], with a comprehensive grid search to

表1：评估了TED、仅大语言模型、仅小语言模型和LLM+SLM方法的性能。基线方法的全部结果（除GPT-4o-mini和ChatEval外）均来自Hu等人 [15]。macF1得分和准确率的前两个结果被突出显示，每个类别中最高结果加粗，第二高结果加下划线。加粗数字表示基线模型的改进具有统计学意义（p值<0小于0.01的t检验）。

模型		ARG-EN				ARG-CN			
		macF1	Acc.	$F1_{真实}$	$F1_{虚假}$	macF1	Acc.	$F1_{真实}$	$F1_{虚假}$
仅大语言模型	GPT-3.5-turbo	0.702	0.813	0.884	0.519	0.725	0.734	0.774	0.676
	GPT-4o-mini	0.691	0.845	0.909	0.472	0.725	0.746	0.780	0.670
仅小语言模型	Bert	0.765	0.862	0.916	0.615	0.753	0.754	0.769	0.737
	EANN	0.763	0.864	0.918	0.608	0.754	0.756	0.773	0.736
	Publisher-Emo	0.766	0.868	0.920	0.611	0.761	0.763	0.784	0.738
	ENDEF	0.768	0.865	0.918	0.618	0.765	0.766	0.779	0.751
大语言模型+小语言模型	BERT + 理由	0.777	0.870	0.921	0.633	0.767	0.769	0.787	0.748
	SuperICL	0.736	0.864	0.920	0.551	0.757	0.759	0.779	0.734
	ARG	0.790	0.878	0.926	0.653	0.784	0.786	0.804	0.764
	ARG-D	0.778	0.870	0.921	0.634	0.771	0.772	0.785	0.756
多智能体	ChatEval(逐个)	0.733	0.860	0.919	0.546	0.694	0.717	0.778	0.611
	ChatEval(同时说话)	0.738	0.869	0.923	0.553	0.694	0.719	0.780	0.608
	TED(我们的)	**0.803**	**0.892**	**0.932**	**0.674**	**0.795**	**0.798**	**0.815**	**0.774**

表2：假新闻检测数据集的统计信息。

	ARG-CN			ARG-EN		
	训练	**Val**	**Test**	**训练**	**Val**	**Test**
Real	2,331	1,172	1,137	2,878	1,030	1,024
Fake	2,873	779	814	1,006	244	234
总计	5,204	1,951	1,951	3,884	1,274	1,258

以确定最佳学习率。性能评估使用测试结果中的最佳验证检查点。对于分析代理，实验使用PyTorch进行，预训练模型使用HuggingFace Transformers库，图神经网络组件使用PyTorch Geometric。文本数据使用适合数据集语言的BERT分词器进行分词。辩论交互使用角色感知嵌入编码，新闻内容使用BERT模型单独编码。此外，每个辩论实例都采用固定图结构，节点代表辩论交互，边捕捉这些交互之间的逻辑流程和关系。边根据辩论序列和角色之间的交互预定义。所有实验都在配备NVIDIA 3090 GPU的机器上运行以加速训练。具体的实验实现细节可以在我们的代码中找到。

4.2 主要结果

表1展示了我们提出的TED框架以及各种其他模型（如仅大语言模型、仅小语言模型和LLM+SLM方法）在ARG-EN和ARG-CN数据集上的性能表现。从结果中，我们观察到：(1) TED表现优于所有其他方法，在所有数据集上均实现了最高的macF1分数和准确率，展现出其在检测假新闻方面的卓越能力。这突显了我们多智能体辩论方法在彻底分析和判断新闻真实性方面的有效性。(2) 仅大语言模型方法，在不进行结构化论证的情况下利用大语言模型，其表现劣于TED。这强调了当大语言模型单独部署时其能力的局限性，缺乏TED中提供的结构化交互和话语机制。(3) 仅小语言模型方法之间的比较显示，每个变体都有渐进式的改进，但均未达到TED的性能水平。这表明虽然传统小语言模型有能力，但它们仍不及TED中多智能体系统提供的稳健和全面的分析框架。(4) LLM+SLM方法，试图结合大语言模型和小语言模型的能力，表现出不同的性能。例如，ARG模型显示出相当的有效性，但仍未达到TED的水平。这表明虽然大语言模型和小语言模型的集成提供了好处，但TED的结构化辩论格式对于最佳性能至关重要。

此外，为了评估我们的TED框架在假新闻检测方面的有效性，我们将其与使用<样式 id='2'>逐个</样式>和<样式 id='4'>同步对话</样式>策略进行多智能体辩论的ChatEval框架 [3], 进行了比较。将这些策略应用于我们的假新闻检测任务，如表1所示，TED在所有指标上始终优于ChatEval。TED的正式辩论结构促进了智能体之间更深入的交互推理，从而允许对新闻内容进行更深入的分析。相比之下，ChatEval的策略在假新闻检测方面未能达到同样的深度。这些结果突出了TED在提高多智能体系统在假新闻检测中的准确性和可靠性方面的优越性。通过培养结构化和动态的交互，TED不仅提高了检测准确率，还确保了对新闻项更彻底的审查。

Table 1: The performance of the TED, as well as the LLM-only, SLM-only, and LLM+SLM methods was assessed. All results (except for those of `GPT-4o-mini` and ChatEval) for the baselines are taken from Hu et al. [15]. The top two results for both macF1-score and accuracy are highlighted, with the highest in each category being bolded and the second highest underlined. Numbers in bold mean that the improvements to the baseline models are statistically significant (t-test with p-value<0.01).

Model		ARG-EN				ARG-CN			
		macF1	Acc.	$F1_{real}$	$F1_{fake}$	macF1	Acc.	$F1_{real}$	$F1_{fake}$
LLM-Only	GPT-3.5-turbo	0.702	0.813	0.884	0.519	0.725	0.734	0.774	0.676
	GPT-4o-mini	0.691	0.845	0.909	0.472	0.725	0.746	0.780	0.670
SLM-Only	Bert	0.765	0.862	0.916	0.615	0.753	0.754	0.769	0.737
	EANN	0.763	0.864	0.918	0.608	0.754	0.756	0.773	0.736
	Publisher-Emo	0.766	0.868	0.920	0.611	0.761	0.763	0.784	0.738
	ENDEF	0.768	0.865	0.918	0.618	0.765	0.766	0.779	0.751
LLM+SLM	Bert + Rationale	0.777	0.870	0.921	0.633	0.767	0.769	0.787	0.748
	SuperICL	0.736	0.864	0.920	0.551	0.757	0.759	0.779	0.734
	ARG	0.790	0.878	0.926	0.653	0.784	0.786	0.804	0.764
	ARG-D	0.778	0.870	0.921	0.634	0.771	0.772	0.785	0.756
Multi-Agents	ChatEval(One-by-One)	0.733	0.860	0.919	0.546	0.694	0.717	0.778	0.611
	ChatEval(Simultaneous-Talk)	0.738	0.869	0.923	0.553	0.694	0.719	0.780	0.608
	TED(Ours)	**0.803**	**0.892**	**0.932**	**0.674**	**0.795**	**0.798**	**0.815**	**0.774**

Table 2: Statistics of the fake news detection datasets.

	ARG-CN			ARG-EN		
	Train	**Val**	**Test**	**Train**	**Val**	**Test**
Real	2,331	1,172	1,137	2,878	1,030	1,024
Fake	2,873	779	814	1,006	244	234
Total	5,204	1,951	1,951	3,884	1,274	1,258

determine the optimal learning rate. Performance is evaluated using the best-validation checkpoint from the testing results. For the Analysis Agent, the experiments were conducted using PyTorch, with the HuggingFace Transformers library for pre-trained models and PyTorch Geometric for graph neural network components. Text data was tokenized using the BERT tokenizer appropriate for the dataset's language. Debate interactions were encoded with role-aware embeddings, and news content was separately encoded using the BERT model. Additionally, a fixed graph structure was employed for each debate instance, with nodes representing debate interactions and edges capturing the logical flow and relationships between these interactions. Edges were predefined based on the debate sequence and interactions between roles. All experiments were run on machines equipped with NVIDIA 3090 GPUs to accelerate training. The specific experimental implementation details can be found in our code.

4.2 Main Results

Table 1 presents the performance of our proposed TED framework and various other models, such as LLM-only, SLM-only, and LLM+SLM methods, on the ARG-EN and ARG-CN datasets. From the results, we observe: (1) TED outperforms all other methods, achieving the highest macF1 scores and accuracy across the datasets, showcasing its exceptional capability in detecting fake news. This underlines the efficacy of our multi-agent debate method in thoroughly analyzing and judging the truth of news items. (2) The LLM-only method, which utilizes LLMs without structured argumentation, exhibits inferior performance compared to TED. This emphasizes the limited capability of LLMs when deployed in isolation, lacking the structured interaction and discourse mechanisms provided in TED. (3) Comparisons among SLM-only methods reveal incremental improvements across each variant, yet none achieve the performance level of TED. This suggests that while traditional SLMs are capable, they fall short of the robust and comprehensive analytical framework offered by the multi-agent system in TED. (4) The LLM+SLM methods, attempting to combine the capabilities of LLMs with SLMs, show varied performance. For instance, the ARG model demonstrates considerable effectiveness but still does not reach the level of TED. This indicates that while the integration of LLM and SLM provides benefits, the structured debate format of TED is crucial for optimal performance.

In addition, to evaluate the effectiveness of our TED framework for fake news detection, we compared it with the ChatEval framework [3], which uses *One-by-One* and *Simultaneous-Talk* strategies for multi-agent debates. Applying these strategies to our fake news detection task, as shown in Table 1, TED consistently outperforms ChatEval across all metrics. The formal debate structure of TED promotes more interactive reasoning among agents, allowing for a deeper analysis of news content. In contrast, ChatEval's strategies do not achieve the same depth in fake news detection. These results highlight TED's superiority in enhancing the accuracy and reliability of multi-agent systems for fake news detection. By fostering structured and dynamic interactions, TED not only improves detection accuracy but also ensures a more thorough examination of the news item.

表3：我们的模型TED在ARG-EN和ARG-CN数据集上的消融研究。粗体数字表示消融模型的改进在统计上显著（t检验，p值<0。01）。

	ARG-EN				ARG-CN			
	macF1	Acc.	$F1_{真实}$	$F1_{虚假}$	macF1	Acc.	$F1_{真实}$	$F1_{虚假}$
完整模型（TED）	**0.803**	**0.892**	**0.932**	**0.674**	**0.795**	**0.798**	**0.815**	**0.774**
无*DebateFlow*代理	0.780	0.874	0.918	0.642	0.767	0.781	0.795	0.739
无合成代理	0.792	0.881	0.924	0.660	0.771	0.787	0.801	0.741
无分析代理	0.729	0.856	0.915	0.543	0.737	0.736	0.785	0.689

5 分析与讨论

5.1 消融研究

我们进行了一项消融研究，以评估TED框架中每个组件的影响。结果表明，移除DebateFlow代理会导致ARG-EN上的macF1分数降低约2.3%，ARG-CN上的macF1分数降低约2.8%。这种下降突出了结构化辩论过程在促进多视角分析方面的重要性，这对于准确确定新闻项的真实性至关重要。移除合成代理会导致macF1分数适度下降，ARG-EN上下降1.1%，ARG-CN上下降2.4%。这突出了合成代理在将辩论交互整合为连贯摘要中的关键作用，从而增强了模型整合不同论点进行精确分类的能力。省略分析代理会导致macF1分数最大幅度的下降，ARG-EN上下降8.0%，ARG-CN上下降4.5%，强调了其在将辩论见解与新闻内容整合中的关键功能。总体而言，消融结果证实TED框架的每个组件对于在假新闻检测中实现高性能都是不可或缺的。DebateFlow代理支持强大的多智能体交互；合成代理确保了辩论结果的有效整合，而分析代理将这些见解与新闻项整合以获得精确结果。

5.2 在不同主干网络上的性能

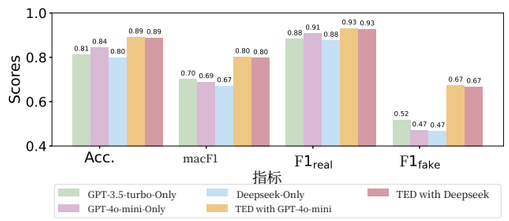


图3：我们的TED框架使用GPT-4o- mini和Deepseek作为闭源主干网络时的性能。

5.2.1 在闭源模型上的性能。 为评估TED框架的鲁棒性和通用性，我们在ARG-EN数据集上进行了实验，通过用不同的闭源模型替换DebateFlow代理和合成代理，具体为GPT-4o-mini和Deepseek[2]。图3表明，与基线LLM-Only模型相比，TED在所有评估指标上始终提升性能，无论使用哪种特定的闭源模型。这一结果突显了该框架对不同大规模模型的适应性，表明TED能有效利用各种闭源主干模型的优势。此外，我们观察到TED with GPT-4o-mini和TED with Deepseek的性能相当，GPT-4o-mini略有优势。这表明虽然这两个模型具有竞争力，但GPT-4o-mini在某些方面似乎略有优势，特别是在处理复杂推理任务时。尽管如此，两种主干模型的结果都强调了TED在提升假新闻检测性能方面的鲁棒性，无论使用哪种特定的闭源模型。这些发现表明TED不仅限于单一主干，而且能有效提升不同闭源模型的表现。TED with GPT-4o-mini的轻微优势进一步证实了TED能够利用每个模型推理能力的固有优势。这种适应性确保了TED在实际应用中的有效性和可靠性。

总体而言，TED框架被证明既有效又灵活，在各种闭源模型上都能保持强劲性能。这些结果验证了TED在不同应用场景中的实用性，使其成为假新闻检测领域的高效工具。

表4：TED使用Qwen2.5和Llama3.1系列模型作为主干网络时的性能。带下划线的数字表示使用开源模型的最佳性能，且具有统计学显著性（p值<0.01的t检验）。

骨干网络	#参数	macF1	Acc	$F1_{real}$	$F1_{fake}$
GPT-4o-mini	-	0.803	0.892	0.932	0.674
Qwen 2.5 [48]	7B	**0.794**	**0.887**	**0.926**	**0.661**
Llama 3.1 [10]	8B	0.787	0.879	0.920	0.653
Llama 3.1 [10]	70B	0.790	0.884	0.923	0.657

5.2.2 在开源模型上的性能。 为了进一步评估TED框架的适应性和鲁棒性，我们使用多个开源语言模型在ARG-EN数据集上测试了其性能，包括Qwen-2.5 [48] 和Llama 3.1系列 [10]。如表4所示，TED始终如一地提升了

[2]https://www.deepseek.com/

Table 3: Ablation study of our model TED on ARG-EN and ARG-CN datasets. Numbers in bold mean that the improvements to the ablation models are statistically significant (t-test with p-value<0.01).

	ARG-EN				ARG-CN			
	macF1	Acc.	$F1_{real}$	$F1_{fake}$	macF1	Acc.	$F1_{real}$	$F1_{fake}$
Full Model (TED)	**0.803**	**0.892**	**0.932**	**0.674**	**0.795**	**0.798**	**0.815**	**0.774**
w/o DebateFlow Agents	0.780	0.874	0.918	0.642	0.767	0.781	0.795	0.739
w/o Synthesis Agent	0.792	0.881	0.924	0.660	0.771	0.787	0.801	0.741
w/o Analysis Agent	0.729	0.856	0.915	0.543	0.737	0.736	0.785	0.689

5 Analysis and Discussion

5.1 Ablation Study

We conducted an ablation study to assess the impact of each component in the TED framework. The results indicate that removing the DebateFlow Agents reduces macF1 scores by approximately 2.3% on ARG-EN and 2.8% on ARG-CN. This decline underscores the importance of the structured debate process in facilitating comprehensive multi-perspective analysis, which is essential for accurately determining the authenticity of news items. Removing the Synthesis Agent results in a moderate decrease in macF1 scores, dropping by 1.1% on ARG-EN and 2.4% on ARG-CN. This highlights the critical role of the Synthesis Agent in consolidating debate interactions into coherent summaries, thereby enhancing the model's ability to integrate diverse arguments effectively for precise classification. Omitting the Analysis Agent causes the most significant drop, with macF1 scores falling by 8.0% on ARG-EN and 4.5% on ARG-CN, emphasizing its crucial function in integrating debate insights with the news content. Overall, the ablation results confirm that each component of the TED framework is indispensable for achieving high performance in fake news detection. The DebateFlow Agents enable robust multi-agent interactions; the Synthesis Agent ensures effective consolidation of debate outcomes, and the Analysis Agent integrates these insights with the news item to get precise results.

5.2 Performance on Different Backbones

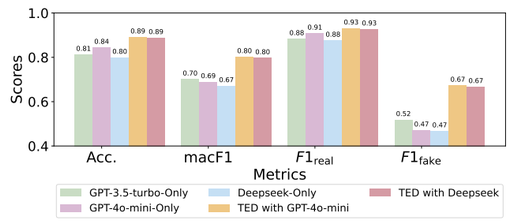


Figure 3: Performance of our framework TED with GPT-4o-mini and Deepseek as closed-source backbones respectively.

5.2.1 Performance on closed-source models. To evaluate the robustness and versatility of the TED framework, we conducted experiments on the ARG-EN dataset by replacing the DebateFlow Agents and Synthesis Agent with different closed-source models, namely GPT-4o-mini and Deepseek[2]. Figure 3 demonstrates that TED consistently enhances performance compared to the baseline LLM-Only model across all evaluation metrics, regardless of the specific closed-source model used. This outcome highlights the framework's adaptability to different large-scale models, showing that TED effectively leverages the unique strengths of various closed-source backbones. Additionally, we observed that the performance of TED with GPT-4o-mini and TED with Deepseek was comparable, with a slight advantage in favor of GPT-4o-mini. This suggests that while the two models are competitive, GPT-4o-mini appears to have a slight edge in some aspects, particularly in handling complex reasoning tasks. Despite this, the results across both backbones emphasize TED's robustness in boosting fake news detection performance regardless of the specific closed-source model used. These findings underline that TED is not limited to a single backbone and can effectively enhance performance across different closed-source models. The slight superiority of TED with GPT-4o-mini reinforces the notion that TED can leverage the inherent strengths of each model's reasoning abilities. This adaptability ensures that TED remains effective and reliable in real-world applications.

Overall, the TED framework proves to be both effective and adaptable, maintaining strong performance across a variety of closed-source models. These results validate TED's utility in diverse application scenarios, making it a highly versatile tool for fake news detection.

Table 4: Performance of TED with Qwen2.5 and Llama3.1 family models as backbones, respectively. Numbers with underlined mean the best performance with an open-source model and are statistically significant(t-test with p-value<0.01).

Backbones	#Param	macF1	Acc	$F1_{real}$	$F1_{fake}$
GPT-4o-mini	-	0.803	0.892	0.932	0.674
Qwen 2.5 [48]	7B	**0.794**	**0.887**	**0.926**	**0.661**
Llama 3.1 [10]	8B	0.787	0.879	0.920	0.653
Llama 3.1 [10]	70B	0.790	0.884	0.923	0.657

5.2.2 Performance on open-source models. To further evaluate the adaptability and robustness of the TED framework, we tested its performance on the ARG-EN dataset using several open-source language models, including Qwen-2.5 [48] and the Llama 3.1 family [10]. As demonstrated in Table 4, TED consistently improves

[2]https://www.deepseek.com/

所有开源模型上的性能，突显了它能够显著增强假新闻检测的能力，而不管所使用的具体语言模型是什么。这表明TED不受骨干模型可用性或类型的限制，使其成为适用于各种实际应用的通用且有效的工具。此外，TED与Qwen 2.5的性能略优于Llama 3.1变体。尽管性能差异很小，但这表明Qwen 2.5特别适合TED的辩论驱动框架。它可能更好地处理复杂的论证和上下文，这对于TED所依赖的结构化评估至关重要。另一方面，TED与Llama 3.1模型的性能也表现出色，所有配置都优于基线方法，进一步验证了TED在不同模型上的有效性。尽管与开源模型的性能低于使用GPT-4o-mini获得的结果，但这很可能归因于模型底层推理能力的差异。开源模型虽然功能强大，但可能不具备与GPT-4o-mini等闭源模型相同的细微推理水平。尽管如此，TED即使使用这些开源骨干模型也能持续产生强劲的结果，证明它仍然能够利用每个模型的优势，以实现高水平的准确率和可靠的检测。

总体而言，研究结果突出了TED的鲁棒性和多功能性。它始终提升了假新闻检测性能。这些发现表明，TED非常适合在各种环境中部署，包括无法访问特定模型或更倾向于使用开源模型的情况。

5.3 案例研究

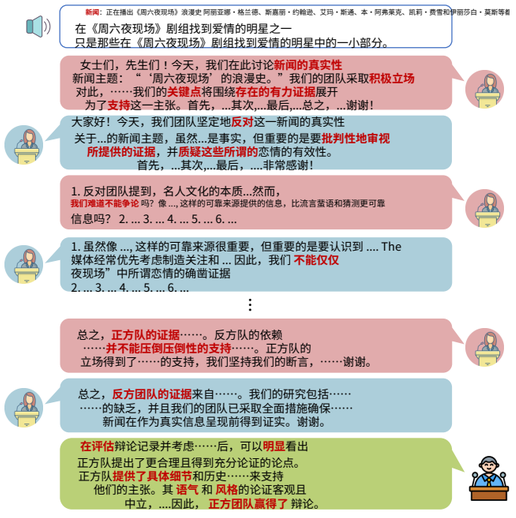


图4：区域辩论案例研究及其辩论报告，可作为可解释假新闻检测的依据。

图4以假新闻辩论为例，呈现了一项比较微观的分析。在该辩论中，参与者被分为支持方和反对方团队，通过开场、交叉询问和总结陈词来完成辩论场景。辩论开始时，支持方明确表示，“本次辩论的主要目的是讨论新闻的真实性”，并提出了关键论点作为证据来支持其立场。反对方采取了类似的方法，提出了他们的主张。在提问和反驳阶段，双方互相挑战对方的观点，并对提出的疑问作出回应。在总结陈词阶段，支持方回顾并总结了之前的辩论，质疑了反对方的论点，并强化了自己的观点。尽管反对方重申了他们的立场，但他们并没有深刻挑战支持方的观点。最终，根据辩论的细节、语气和风格等因素，评委决定支持支持方。整个辩论过程形成了一个“辩论流程”，清晰地展示了推理过程。评委的辩论报告也作为判断新闻是否为假的基础，从而实现了可解释的假新闻检测。这种可解释性对于现实生活中的信息验证至关重要，代表了TED框架的一个显著优势，这也是其他模型难以实现的一个方面。

6 结论

总之，我们的研究介绍了并评估了TED框架，这是一种结合了大语言模型优势的结构化多智能体辩论的新型假新闻检测方法。通过在ARG-EN和ARG-CN数据集上进行广泛的实验验证，TED在多个性能指标上始终优于传统的机器学习和独立的大语言模型。受人类辩论的启发，该框架的设计利用了不同的视角和结构化论证，为评估新闻的真实性提供了一种稳健的机制。我们的工作不仅通过提高准确性和可解释性推动了假新闻检测领域的发展，还为开发负责任的信息系统提供了宝贵的见解。通过实现更可靠和透明的内容评估，TED有助于创建能够更好地支持用户在日益复杂的数字信息环境中导航和评估的信息检索系统。

附录A

在ARG-CN数据集的实验中，我们使用的提示是上述提示集的对应中文版本。其他所有功能提示都可以在我们的代码文件中找到：https://github.com/LiuYuHan31/TED_虚假新闻检测。

致谢

这项工作得到了北京优秀青年科学家计划（项目编号：BJJWZYJH012019100020098）和中国人民大学“双一流”建设重大创新与规划跨学科平台——智能社会治理平台的资助。刘宇涵由中国人民大学的“求是学者-东良”项目（项目编号：RUC24QSDL015）提供支持。

performance across all the open-source models, highlighting its ability to significantly enhance fake news detection regardless of the specific language model used. This shows that TED is not constrained by the availability or type of backbone model, making it a versatile and effective tool for various real-world applications. Additionally, TED's performance with Qwen 2.5 slightly outperforms the Llama 3.1 variants. Although the differences in performance are minimal, this suggests that Qwen 2.5 is particularly well-suited to TED's debate-driven framework. It may offer a better handling of complex argumentation and context, which is essential for the structured evaluation TED relies on. On the other hand, TED with the Llama 3.1 models also performs admirably, and all configurations lead to better results than the baseline methods, further validating TED's effectiveness across different models. Although the performance with open-source models is lower than the results obtained with GPT-4o-mini, this can likely be attributed to differences in the underlying reasoning capabilities of the models. Open-source models, while highly capable, may not exhibit the same level of nuanced reasoning as closed-source models like GPT-4o-mini. Nevertheless, TED continues to produce strong results even with these open-source backbones, demonstrating that it can still leverage each model's strengths to achieve high levels of accuracy and reliable detection.

Overall, the results highlight TED's robustness and versatility. It consistently enhances fake news detection performance. These findings suggest that TED is well-suited for deployment in various settings, including scenarios where access to specific models may be limited or open-source models are preferred.

5.3 Case Study

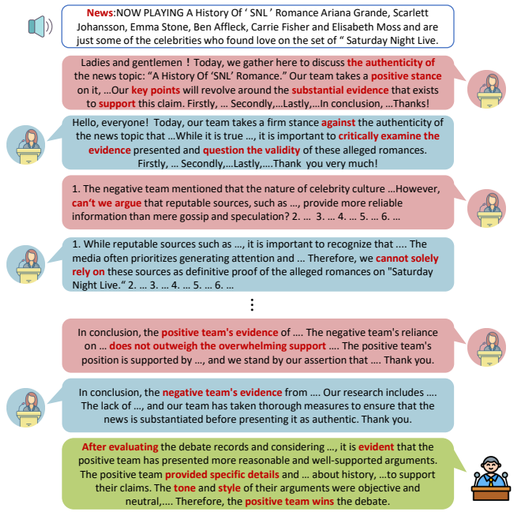


Figure 4: A real debate case study, with its debate report, can serve as justification for interpretable fake news detection.

Figure 4 presents a comparative micro-analysis of a debate process on fake news as a case study. Participants are divided into pro and con teams in this debate, completing the debate scenario through opening, cross-examinations, and closing statements. At the beginning of the debate, the pro team clearly stated, “The primary objective of this debate is to discuss the truth of the news," and they presented key points as evidence to support their stance. The con team adopted a similar approach, presenting their assertions. During the Questioning and Rebuttal phase, both sides challenged each other's viewpoints and responded to the queries raised. In the Closing Statement phase, the pro team reviewed and summarized the prior debate questioned the con team's arguments and reinforced their own points. Although the con team reiterated their position, they did not profoundly challenge the pro team's viewpoints. Ultimately, based on factors such as the debate's detail, tone, and style, the judges decided to favor the pro team. This entire debate process forms a “DebateFlow," clearly demonstrating the reasoning process. The judges' debate report also serves as a basis for determining whether news is fake, thus achieving an interpretable fake news detection. Such interpretability is crucial for real-life information verification, representing a significant advantage of the TED framework, an aspect that other models struggle to achieve.

6 Conclusion

In conclusion, our study introduces and evaluates the TED framework, a novel approach to fake news detection that incorporates the strengths of LLMs within a structured multi-agent debate. Through extensive experimental validation on the ARG-EN and ARG-CN datasets, TED consistently outperformed traditional machine learning and standalone LLMs across multiple performance metrics. Inspired by human debate, the framework's design leverages diverse perspectives and structured argumentation, providing a robust mechanism for assessing the truth of news items. Our work not only advances the field of fake news detection by enhancing accuracy and interpretability but also offers valuable insights for developing responsible information systems. By enabling more reliable and transparent content evaluation, TED contributes to creating information retrieval systems that can better support users in navigating and assessing the growing complexity of digital information.

A Appendix

In the experiment using the ARG-CN dataset, the prompt we used was the corresponding Chinese version of the aforementioned prompt set. And all other function prompts can be found in our code files at https://github.com/LiuYuHan31/TED_fake-news-detction.

Acknowledgments

This work was supported by the Beijing Outstanding Young Scientist Program (NO. BJJWZYJH012019100020098) and the Intelligent Social Governance Platform, a Major Innovation & Planning Interdisciplinary Platform for the “Double-First Class” Initiative at Renmin University of China. Yuhan Liu is supported by the "Qiushi Academic-Dongliang" Project of Renmin University of China (No. RUC24QSDL015).

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